

Supplementary Material for “Construction of Synchronous Machine Surrogates from Limited Measurements: A Physics-Prior Neural ODE Approach”

Ruizhi Yu, Wei Gu, Shuai Lu, Ke Wang, Suhan Zhang

This document supplements the main text. Section S1 lists the synchronous machine models of the family in full. Section S2 proves Proposition 1 of the main text. Section S3 gives the gradient of the port-only loss. Section S4 gives the numerical differentiation formula and its residual equation. Section S5 lists the parameters of the IEEE 14-bus case study. Section S6 lists the training settings and the sample counts of the case studies. Equation, section, table, and proposition numbers without the prefix S refer to the main text.

S1. THE SYNCHRONOUS MACHINE MODEL FAMILY

A. The Sauer–Pai Model with GENROU Saturation

The sixth-order Sauer–Pai model [1] with the GENROU saturation [2] reads

$$V_d = V_h \sin(\delta - \theta_h), \quad V_q = V_h \cos(\delta - \theta_h), \quad (\text{S1a})$$

$$\dot{\delta} = \Omega_b(\omega - 1), \quad 2H\dot{\omega} = T_m - T_e - D(\omega - 1), \quad (\text{S1b})$$

$$0 = V_d + R_a I_d - E_d'' - X_q'' I_q, \quad (\text{S1c})$$

$$0 = V_q + R_a I_q - E_q'' + X_d'' I_d, \quad (\text{S1d})$$

$$T_e = E_d'' I_d + E_q'' I_q + (X_q'' - X_d'') I_d I_q, \quad (\text{S1e})$$

$$T_{d0}' \dot{E}_q' = -E_q' - (X_d - X_d') (I_d - \gamma_d \eta_d) + E_{fd} - S_e E_q'', \quad (\text{S1f})$$

$$T_{q0}' \dot{E}_d' = -E_d' + (X_q - X_q') (I_q - \gamma_q \eta_q) - S_e g_{qd} E_d'', \quad (\text{S1g})$$

$$T_{d0}'' \dot{\psi}_{1d} = -\psi_{1d} + E_q' - (X_d' - X_l) I_d, \quad (\text{S1h})$$

$$T_{q0}'' \dot{\psi}_{2q} = -\psi_{2q} - E_d' - (X_q' - X_l) I_q, \quad (\text{S1i})$$

$$E_q' = a_d E_q'' + b_d \psi_{1d}, \quad E_d'' = a_q E_d' - b_q \psi_{2q}, \quad (\text{S1j})$$

$$S_e = \frac{B_s(\psi_a - A_s)^2}{\psi_a}, \quad \psi_a = \sqrt{(E_d'')^2 + (E_q'')^2}. \quad (\text{S1k})$$

Equations (S1a)–(S1e) and (S1j) are the swing and stator equations (3) of the main text, and the symbols there keep their meaning. The rotor equations (S1f)–(S1i) take the transient EMFs E_q' , E_d' and the damper flux linkages ψ_{1d} , ψ_{2q} as rotor states, where E_{fd} is the field voltage, X_d , X_q the synchronous reactances, X_d' , X_q' the transient reactances, X_l the leakage reactance, and T_{d0}' , T_{q0}' , T_{d0}'' , T_{q0}'' the open-circuit time constants. The coupling terms carry the sub-transient leakage fluxes

$$\eta_d = \psi_{1d} + (X_d' - X_l) I_d - E_q', \quad \eta_q = \psi_{2q} + (X_q' - X_l) I_q + E_d', \quad (\text{S2})$$

and the constant coefficients

$$\gamma_d = \frac{X_d' - X_d''}{(X_d' - X_l)^2}, \quad \gamma_q = \frac{X_q' - X_q''}{(X_q' - X_l)^2}. \quad (\text{S3})$$

The EMF map (S1j) has the coefficients

$$a_d = \frac{X_d'' - X_l}{X_d' - X_l}, \quad b_d = \frac{X_d' - X_d''}{X_d' - X_l}, \quad (\text{S4})$$

$$a_q = \frac{X_q'' - X_l}{X_q' - X_l}, \quad b_q = \frac{X_q' - X_q''}{X_q' - X_l}.$$

The saturation factor (S1k) is a quadratic function of the air-gap flux magnitude ψ_a that vanishes for $\psi_a \leq A_s$, and it enters the q axis through $g_{qd} = (X_q - X_l)/(X_d - X_l)$. Its coefficients A_s and B_s are fitted so that S_e passes through the standard saturation points (1.0, $S_{1.0}$) and (1.2, $S_{1.2}$) [2], and $S_e = 0$ recovers the unsaturated Sauer–Pai model.

B. The Marconato and the Anderson–Fouad Models

The Marconato model [3] takes the sub-transient EMFs themselves as rotor states, E_q' , E_d' , E_q'' , E_d'' , and its rotor equations read

$$T_{d0}' \dot{E}_q' = -E_q' - (X_d - X_d' - \gamma_d^M) I_d + \left(1 - \frac{T_{AA}}{T_{d0}'}\right) E_{fd}, \quad (\text{S5a})$$

$$T_{q0}' \dot{E}_d' = -E_d' + (X_q - X_q' - \gamma_q^M) I_q, \quad (\text{S5b})$$

$$T_{d0}'' \dot{E}_q'' = -E_q'' + E_q' - (X_d' - X_d'' + \gamma_d^M) I_d + \frac{T_{AA}}{T_{d0}'} E_{fd}, \quad (\text{S5c})$$

$$T_{q0}'' \dot{E}_d'' = -E_d'' + E_d' + (X_q' - X_q'' + \gamma_q^M) I_q, \quad (\text{S5d})$$

where T_{AA} is the additional leakage time constant of the d axis and

$$\gamma_d^M = \frac{T_{d0}'' X_d''}{T_{d0}' X_d'} (X_d - X_d'), \quad \gamma_q^M = \frac{T_{q0}'' X_q''}{T_{q0}' X_q'} (X_q - X_q'). \quad (\text{S6})$$

The sub-transient EMFs are states, so the EMF map (S1j) is the identity, and the stator and torque equations (S1c)–(S1e) apply unchanged. The Anderson–Fouad model [3] is (S5) with $\gamma_d^M = \gamma_q^M = 0$ and $T_{AA} = 0$, that is, four cascaded first-order equations without coupling.

C. The Reduced-Order Models

The reduced-order members drop the rotor states whose time constants are separated from the electromechanical time scale [1], [3]. The two-axis model drops the damper windings, whose time constants T''_{d0} and T'''_{q0} are of the order of tens of milliseconds, and keeps

$$T'_{d0}\dot{E}'_q = -E'_q - (X_d - X'_d)I_d + E_{fd}, \quad (\text{S7a})$$

$$T'_{q0}\dot{E}'_d = -E'_d + (X_q - X'_q)I_q, \quad (\text{S7b})$$

$$0 = V_d + R_a I_d - E'_d - X'_q I_q, \quad (\text{S7c})$$

$$0 = V_q + R_a I_q - E'_q + X'_d I_d, \quad (\text{S7d})$$

$$T_e = E'_d I_d + E'_q I_q + (X'_q - X'_d)I_d I_q, \quad (\text{S7e})$$

so the transient EMFs act behind the transient reactances. The one-axis model further drops E'_d and keeps

$$T'_{d0}\dot{E}'_q = -E'_q - (X_d - X'_d)I_d + E_{fd}, \quad (\text{S8a})$$

$$0 = V_d + R_a I_d - X_q I_q, \quad (\text{S8b})$$

$$0 = V_q + R_a I_q - E'_q + X'_d I_d, \quad (\text{S8c})$$

$$T_e = E'_q I_q + (X_q - X'_d)I_d I_q. \quad (\text{S8d})$$

The classical model holds a constant EMF E' behind the transient reactance X'_d , since T'_{d0} is of the order of seconds and the field flux barely changes within a first-swing study, and keeps

$$0 = V_d + R_a I_d - X'_d I_q, \quad (\text{S9a})$$

$$0 = V_q + R_a I_q - E' + X'_d I_d, \quad (\text{S9b})$$

$$T_e = E' I_q. \quad (\text{S9c})$$

In every member the Park transformation (S1a) and the swing equations (S1b) are unchanged.

D. Saturation Conventions

Even for the saturation no consensus exists. Reference [1] distributes one open-circuit saturation function of the air-gap flux over the field and damper windings, which the GENROU form (S1f)–(S1g) and (S1k) follows, while [3] admits a separate saturation function per winding, realized as a three-region piecewise curve or as a polynomial fit of the saturation characteristic. The rotor equations thus vary across references, machine types, and model orders, which is why the main text learns the rotor model instead of presuming one member.

S2. PROOF OF PROPOSITION 1

Proposition 1 of the main text bounds every trajectory of the learned rotor model (7), for any network weights, by (10).

Proof. Write (7) as $\dot{\mathbf{x}}_r = \mathbf{\Lambda} \mathbf{x}_r + \mathbf{p}(t)$ with $\mathbf{\Lambda} = \mathbf{T}^{-1} \mathbf{A}$ and $\mathbf{p} = \mathbf{T}^{-1}(\mathbf{B} \mathbf{u}_r + \mathbf{r} \theta)$. The output bound of (9) gives $\|\mathbf{r} \theta_i\| \leq c_i$, so $\|\mathbf{p}(t)\|_2 \leq \bar{u} + \|\mathbf{T}^{-1} \mathbf{c}\|_2$ under the input bound $\|\mathbf{T}^{-1} \mathbf{B} \mathbf{u}_r(t)\|_2 \leq \bar{u}$. Every eigenvalue of $\mathbf{\Lambda}$ is negative, so constants $\gamma \geq 1$ and $\lambda > 0$ with $\|e^{\mathbf{\Lambda} t}\|_2 \leq \gamma e^{-\lambda t}$ exist [4]. The variation-of-constants formula gives $\mathbf{x}_r(t) = e^{\mathbf{\Lambda} t} \mathbf{x}_r(0) + \int_0^t e^{\mathbf{\Lambda}(t-\tau)} \mathbf{p}(\tau) d\tau$. Bounding the integrand by $\gamma e^{-\lambda(t-\tau)} \|\mathbf{p}\|_2$ and integrating yields (10). $\square$

S3. GRADIENT OF THE PORT-ONLY LOSS

The gradient of the loss (14) is obtained by first discretizing the forward simulation and then differentiating the discrete solution, which is exactly what reverse-mode automatic differentiation computes. Let $\mathbf{y} = [\delta, \omega, \mathbf{x}_r^\top]^\top$ collect the surrogate state and let Φ_n denote one step of a fixed-step fourth-order Runge–Kutta scheme applied to the surrogate dynamics under the recorded inputs. The forward propagation of one record of L steps is

$$\mathbf{y}_0 = \iota(\mathbf{X}, \mathbf{P}), \quad \mathbf{y}_{n+1} = \Phi_n(\mathbf{y}_n; \mathbf{P}), \quad n = 0, \dots, L-1, \quad (\text{S10})$$

where ι is the initial-condition map of Section III-C of the main text, $\mathbf{X}$ are the latent initial angles, and $\mathbf{P}$ are the shared parameters. For the backward propagation, write ℓ_n for the contribution of sample n to (14), and $\mathbf{a}_n = \partial \mathcal{L} / \partial \mathbf{y}_n$ for the sensitivity of the loss to the state at sample n . The sensitivity obeys the reverse recursion

$$\mathbf{a}_L = \frac{\partial \ell_L}{\partial \mathbf{y}_L}, \quad \mathbf{a}_n = \frac{\partial \ell_n}{\partial \mathbf{y}_n} + \left(\frac{\partial \Phi_n}{\partial \mathbf{y}_n} \right)^\top \mathbf{a}_{n+1} \quad (\text{S11})$$

for $n = L-1, \dots, 0$. The gradient with respect to the learnable variables $\boldsymbol{\vartheta} = (\mathbf{X}, \mathbf{P})$ then assembles from three terms,

$$\frac{d\mathcal{L}}{d\boldsymbol{\vartheta}} = \sum_{n=0}^{L-1} \left(\frac{\partial \Phi_n}{\partial \boldsymbol{\vartheta}} \right)^\top \mathbf{a}_{n+1} + \sum_{n=0}^L \frac{\partial \ell_n}{\partial \boldsymbol{\vartheta}} + \left(\frac{\partial \mathbf{y}_0}{\partial \boldsymbol{\vartheta}} \right)^\top \mathbf{a}_0, \quad (\text{S12})$$

summed over the records. The first term carries the influence of $\mathbf{P}$ on the dynamics, and the second its direct influence on the predicted current through the stator equations (3c)–(3d). The third carries the influence of the initial condition. It is the only term that reaches $\mathbf{X}$, and it adds to the gradient that the first two terms already give $\mathbf{P}$.

S4. THE NUMERICAL DIFFERENTIATION FORMULA

Write $\mathbf{w} = [\mathbf{x}^\top, \mathbf{y}^\top]^\top$ for the full unknown vector of the DAE (1), $\mathbf{F} = [\mathbf{f}^\top, \mathbf{g}^\top]^\top$ for its right-hand side, and $M = \text{diag}(I, 0)$ for the mass matrix, so that the DAE is $M \dot{\mathbf{w}} = \mathbf{F}(t, \mathbf{w})$. The step from t_n to t_{n+1} is taken by the numerical differentiation formula (NDF) of order k [5],

$$M \left(\sum_{m=1}^k \frac{1}{m} \nabla^m \mathbf{w}_{n+1} - \kappa \gamma_k \nabla^{k+1} \mathbf{w}_{n+1} \right) = h \mathbf{F}(t_{n+1}, \mathbf{w}_{n+1}), \quad (\text{S13})$$

where h is the step size, ∇ is the backward difference operator with $\nabla^0 \mathbf{w}_n = \mathbf{w}_n$ and $\nabla^{j+1} \mathbf{w}_n = \nabla^j \mathbf{w}_n - \nabla^j \mathbf{w}_{n-1}$, $\gamma_k = \sum_{j=1}^k 1/j$, and κ is the damping coefficient that lowers the leading truncation term. With the predictor $\mathbf{w}_{n+1}^{(0)} = \sum_{m=0}^k \nabla^m \mathbf{w}_n$, equation (S13) becomes the residual equation (15) of the main text,

$$\mathbf{G}(\mathbf{w}) = M(\mathbf{w} - \mathbf{w}_{n+1}^{(0)} + \boldsymbol{\Psi}) - \alpha_k \mathbf{F}(t_{n+1}, \mathbf{w}) = \mathbf{0}, \quad (\text{S14})$$

$$\alpha_k = \frac{h}{(1 - \kappa) \gamma_k}, \quad \boldsymbol{\Psi} = \frac{1}{(1 - \kappa) \gamma_k} \sum_{m=1}^k \gamma_m \nabla^m \mathbf{w}_n,$$

which is one nonlinear system per step. The simplified Newton iteration that solves it is (16) of the main text.

S5. PARAMETERS OF THE IEEE 14-BUS CASE STUDY

The five members of the family in Section V-A of the main text share the parameters of the machine at bus 2 of the IEEE 14-bus system [3], namely $H = 3.924$ s, $D = 1.2$, $R_a = 0.0052$, $X_d = 1.75$, $X_q = 1.633$, $X'_d = 0.308$, $X'_q = 0.6$, $X''_d = X''_q = 0.217$, and $X_l = 0$ in per unit, and $T'_{d0} = 6.1$, $T'_{q0} = 0.3$, $T''_{d0} = 0.04$, and $T''_{q0} = 0.099$ s. The GENROU setting adds the saturation factors $S_{1.0} = 0.27$ and $S_{1.2} = 1.14$. Every learnable parameter of the surrogate starts from these exact values.

S6. TRAINING SETTINGS AND SAMPLE COUNTS

The correction network (9) has two hidden layers of 64 units with the SiLU activation, and its output layer is zero-initialized. The bound is $c = 1$, raised to $c = 3 \cdot 1$ for the GENROU settings so that the saturation can be absorbed. In the loss (14), the current and the speed are normalized by the standard deviation of the recorded quantity, the weight is $\mu = 1$, and the samples of the fault-on period are weighted by 3. Training uses AdamW for 1500 steps with a learning rate of 6×10^{-3} for the weights and the parameters, 3×10^{-2} for the latent δ_0 , a linear warm-up over the first 5% of the steps, and a cosine decay to 2% of the peak. Each step draws 128 windows of 2.5 s that start at the pre-fault steady state, and the surrogate is integrated by the fixed-step fourth-order Runge–Kutta scheme at the 1 ms sampling interval of the measurements. Table SI lists the sample counts. The clearing times of the IEEE 14-bus study run from 0.10 to 0.70 s in steps of 0.03 s, and those of the NPCC and 2224-bus studies form a grid across the critical clearing time of each scenario. One full-physics run supplies one trajectory for every recorded machine, so the NPCC runs are shared by the twelve surrogated machines and the runs of each 2224-bus family by its trained members.

TABLE SI
SAMPLE COUNTS OF THE CASE STUDIES

Study	Scenarios	Loading levels	Runs
IEEE 14-bus	6	7	882
IEEE 14-bus, GENROU	6	4	504
NPCC	7	5	745
2224-bus	6	1	112

S7. FAULT SCENARIOS OF THE NPCC AND 2224-BUS STUDIES

Table SII lists the fault scenarios of Figs. 5 and 6 of the main text. Every scenario is a three-phase fault at the bus in the column Fault, cleared by tripping the branch between the two buses in the column Branch. The label carries the faulted bus after F and the index of the tripped branch in the case file after L. The branch of F87-L230 is a transformer, and every other branch is a line.

TABLE SII
FAULT SCENARIOS OF THE TWO LARGE SYSTEMS

NPCC			2224-bus		
Scenario	Fault	Branch	Scenario	Fault	Branch
F85-L108	85	85–112	F316-L302	316	316–163
F87-L230	87	87–85	F14-L308	14	14–344
F115-L157	115	115–135	F333-L293	333	333–98
F133-L201	133	133–135	F316-L260	316	316–98
F135-L204	135	135–139	F130-L316	130	130–350
F131-L195	131	131–135	F318-L1427	318	318–356
F132-L197	132	132–133	F318-L1428	318	318–357
			F40-L272	40	40–320

REFERENCES

- [1] P. W. Sauer, M. A. Pai, and J. H. Chow, *Power System Dynamics and Stability: With Synchronphasor Measurement and Power System Toolbox*, 2nd ed. IEEE Press / Wiley, 2018.
- [2] H. Cui, F. Li, and K. Tomsovic, “Hybrid symbolic-numeric framework for power system modeling and analysis,” *IEEE Transactions on Power Systems*, vol. 36, no. 2, pp. 1373–1384, 2021.
- [3] F. Milano, *Power System Modelling and Scripting*. Berlin, Heidelberg: Springer, 2010.
- [4] H. K. Khalil, *Nonlinear Systems*, 3rd ed. Upper Saddle River, NJ: Prentice Hall, 2002.
- [5] R. W. Klopfenstein, “Numerical differentiation formulas for stiff systems of ordinary differential equations,” *RCA Review*, vol. 32, no. 3, pp. 447–462, 1971.